

Chadli Bendjedid University, El Tarf

Faculty Of Science And Technology

Department Of Computer Science

Free Software



Instructor :

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Information about the course:

Teaching Unit: UEM11

Subject: Free Software (Open Source)

Credits: 2

Coefficient: 1

Course Objectives:

Familiarization with computer tools and the Internet.

Recommended Prerequisite Knowledge:

General knowledge in computer science.

Evaluation Method : Exam (100%)

Course Content :**Chapter 1 : Information Technologies****1.1. Definitions****1.2. Tools****1.2.1. Hardware**

1. Computers : desktops, laptops, servers, workstations
2. Communication networks: routers, switches, modems, optical fibers, Wi-Fi
3. Peripherals: printers, scanners, keyboards, mice, cameras, sensors
4. Mobile devices: smartphones, tablets, personal digital assistants (PDAs)
5. Smart chips: RFID, NFC, embedded processors, IoT sensors (Internet of Things)

1.2.2. Software

1. Operating systems: Windows, Linux, macOS, Android, iOS
2. Business applications: ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), human resources, accounting software, etc.
3. Office software: word processing (Word), spreadsheets (Excel), presentations (PowerPoint), collaboration tools (Google Workspace, Microsoft 365)

1.3. Applications

1. Communication spaces: Internet, Intranet, Extranet
2. Multimedia: audio conferencing, video conferencing
3. Electronic Data Interchange (EDI)
4. Workflows

Chapter 2 : Open Source Tools

1. Introduction (history, advantages/disadvantages, and licensing)
 2. Development environment (Introduction to Linux, Introduction to code editors)
 3. Office tools (LibreOffice suite)
 4. Collaboration (storage and sharing)
 5. Contributing to an open-source project
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Chapter 1: IT Information Technology

1. Introduction

Information Technology plays a central role in higher education. It supports teaching, research, and administration. It enables access to digital libraries, online courses, and collaborative platforms.

It improves communication between students, teachers, and institutions. It allows data-driven decision-making and efficient management of academic resources.

IT tools enhance learning experiences through virtual labs, learning management systems, and interactive content. Its impact reshapes how knowledge is delivered, shared, and applied in universities.

2. Definition of Information Technology

1.1 General definition:

Information Technology (IT) is the study, design, development, implementation, support, and management of computer-based information systems.

2.1 Practical definition:

IT refers to the use of computers, software, networks, and electronic systems to process and distribute data and information.

3.1 ISO definition:

According to ISO/IEC, Information Technology covers all technologies for processing, storing, and communicating information.

NB: ISO/IEC stands for **International Organization for Standardization** and **International Electrotechnical Commission**.

3. Importance of IT

- IT helps organizations manage data and information efficiently.
- It improves communication through email, video conferencing, and collaborative tools.
- It supports decision-making with databases and information systems.
- It enables automation of tasks, reducing time and errors.
- It connects people worldwide through the Internet.

4. Main Components of IT

- **Hardware:** Physical devices like computers, servers, routers.
- **Software:** Operating systems, applications, and tools.
- **Networks:** Internet, intranet, wireless systems.
- **Data:** The raw material processed into useful information.
- **People:** Users, administrators, and IT professionals.

5. Information Technology Tools

Information Technology relies on five main components. Each one plays a specific role.

5.1 Hardwar

5.1.1 Computer

A computer is a programmable electronic device that processes, stores, and manipulates data according to instructions stored in its memory. It has two main parts:

- **Hardware** refers to the physical components such as keyboard and mouse
- **Software** refers to the programs and instructions that enable the device to perform tasks

➤ **Basic operation:**

- **Input:** receives data through input devices such as keyboard or mouse.
- **Processing:** executes arithmetic and logical operations.
- **Output:** produces information through display or print.
- **Storage:** saves data for later use

➤ **Main components:**

- **CPU** executes instructions
- **RAM** holds temporary data during processing
- **Hard drive or SSD** provides permanent data storage
- **GPU** manages graphics rendering
- **Motherboard** connects all internal components

➤ **Origin of the term**

The English word “computer” comes from the Latin *computare*, meaning “to calculate”
The French term “ordinateur” was introduced in 1955 by Jacques Perret at IBM’s request to describe an information processing machine

- **Types of Computers:** Computers can be classified in different ways: by size, purpose, and processing power.

a. Desktop Computers (PC de bureau)

Explanation: Personal computers designed to be placed on a desk.

Features: Separate monitor, keyboard, mouse, and system unit.

Advantages:

- ▶ Easy to upgrade and repair.
- ▶ More powerful than laptops of the same price.

Use: Home, office, schools, general tasks.



a. Laptop Computers (Ordinateurs portables)

Explanation: Personal computers that integrate screen, keyboard, and battery in one device.

Features: Lightweight, runs on battery or power adapter.

Advantages:

- ▶ *Mobility.*
- ▶ *Saves space.*

Limitations:

- ▶ *More expensive than desktops with the same performance.*
- ▶ *Harder to upgrade or repair.*

Use: Students, business professionals, mobile work.



- **Key Features to Consider When Buying a Computer :**

When you buy a computer, focus on the features that match your needs.

- Check the **processor (CPU)**; it controls speed and performance. Choose at least an Intel i5 or AMD Ryzen 5 for multitasking.
- Look at the **RAM**; 8 GB is the minimum for general use, 16 GB is better for programming or design.
- The **storage type** matters; an **SSD** loads programs faster than an HDD.
- The **graphics card (GPU)** is important for gaming, video editing, and AI tasks.
- The **battery life** affects portability; laptops with longer battery duration are better for study and travel.
- Check the **screen size and resolution** for comfort during long work sessions.

- Ensure it has enough **ports** (USB, HDMI, etc.) and **connectivity options** like Wi-Fi and Bluetooth.
- Finally, consider the **operating system** and your **budget** to find a balanced choice.
- The following table summarizes all the main characteristics you should consider before buying a computer.

Characteristic	Key Points	Recommended Specs by Use Case
Processor (CPU)	Check generation and core count	- Office: Intel i3 / Ryzen 3- Programming: i5 / Ryzen 5- AI or rendering: i7/i9 / Ryzen 7/9
Memory (RAM)	Affects multitasking and performance	- Minimum 8 GB- 16 GB for development or graphic work- 32 GB+ for AI or heavy workloads
Storage	Impacts speed and capacity	- SSD mandatory- 256 GB minimum- 512 GB+ for large files- Optional HDD for extra space
Graphics (GPU)	Needed for games, AI, 3D, video editing	- Integrated for basic tasks- Dedicated with 4 GB VRAM+ for demanding work
Battery (Laptops)	Important for mobility	- Real battery life of 6+ hours recommended
Display	Affects comfort and clarity	- Full HD (1920×1080) minimum- IPS panel for better viewing angles
Connectivity	Determines compatibility with devices	- Enough USB ports- HDMI or DisplayPort- Wi-Fi 5 minimum, Wi-Fi 6 recommended- SD card reader optional
Keyboard & Comfort	Impacts daily use	- Good key feedback- Backlight for low-light work
Operating System	Depends on software needs	- Windows, macOS, or Linux based on your tools
Upgradability	Extends lifespan	- Check if RAM or storage can be expanded
Budget	Guides choices	- Define a clear budget- Compare performance vs price

b. Servers

Explanation: Powerful computers that provide services, resources, or data to other computers (clients) in a network.

Features: High storage, powerful processors, designed for 24/7 operation.

Types: File servers, web servers, database servers.

Use: Websites, cloud services, company networks.

Difference: Unlike PCs, servers are not for personal use but to serve multiple users at the same time.



c. Workstations (Stations de travail)

Explanation: High-performance computers designed for technical or scientific applications.

Features: Faster CPUs, large memory, advanced graphics cards.

Use: Engineering design, 3D modeling, video editing, scientific simulations.

Difference: More powerful than a desktop PC but less expensive and smaller than a mainframe.



Comparaison :

Type	Main Use	Portability	Performance	Users
Desktop PC	General tasks, offices, homes	No	Medium	One
Laptop	Mobile work, students	Yes	Medium	One
Server	Provides resources to a network	No	High	Many
Workstation	Professional technical tasks	No	Very High	One

5.1.2 Communication Networks

Definition: A communication network is a system of connected nodes, such as devices or people that enables information transmission through wired or wireless methods. It can be a computer-based infrastructure for data, voice, and video exchange, or an organizational structure that defines how information flows between individuals and departments. Main types include organizational networks and technical networks such as LANs and WANs, which differ in scale and application

Types of communication networks

1) Organizational networks describe patterns of **person-to-person** information flow within a group or organization:

- ✓ **Formal networks** follow official hierarchical structures and use established channels like memos and emails
- ✓ **Informal networks** arise through social interaction and support information sharing beyond formal structures

2) Computer and technical networks focus on data transmission **between devices**:

- ✓ **LAN** connects devices in a limited area such as offices or homes
- ✓ **WAN** connects multiple LANs across large geographic areas
- ✓ **Wireless networks** use radio waves such as **Wi-Fi** and **Bluetooth** instead of cables

How they function

- Data transmission occurs through physical media such as cables or non-physical media such as radio waves
- Network architecture, often based on the OSI or TCP/IP model, standardizes data formatting, routing, and delivery
- Protocols define the rules for communication between devices, ensuring reliable and efficient data exchange

Importance and applications

- **Connectivity:** links people and devices over any distance
- **Efficiency:** supports collaboration and information flow in organizations
- **Broad applications:** essential in finance, healthcare, IoT, and other fields

a. Routers

Definition: A router is a device that connects different networks together, usually a local network (LAN) to the Internet.

Function: *Directs (routes) data packets between networks.*

Assigns IP addresses to devices.

Provides security features like firewalls and encryption.

Example: The home router that connects your Wi-Fi devices to the Internet.



b. Switches (Commutateurs)

Definition: A switch connects multiple devices inside the same network (LAN).

Function:

- ▶ *Receives data packets and forwards them to the correct device inside the network.*
- ▶ *Reduces collisions and increases network efficiency compared to hubs.*



Types:

- ▶ *Unmanaged switches: simple, plug-and-play.*
- ▶ *Managed switches: advanced, with configuration options for large networks.*

Example: A switch in a university lab connecting all classroom computers.

c. Modems

Definition: A modem (Modulator-Demodulator) converts digital data from a computer into signals that can travel over telephone lines, cable, or fiber.

Function:

Translates signals between the Internet Service Provider (ISP) and your network.

Types:

- ▶ *DSL modem (telephone lines)*
- ▶ *Cable modem (TV cable lines)*
- ▶ *Fiber modem (optical fiber)*

Example: The box provided by your ISP to connect to the Internet.



d. Optical Fibers (Fibres optiques)

Definition: A high-speed transmission medium that uses light signals to carry data.

Features:

- ▶ *Very high bandwidth (up to terabits per second).*
- ▶ *Long-distance communication with minimal signal loss.*
- ▶ *Immune to electromagnetic interference.*

Use: Internet backbone, undersea cables, modern ISPs.

Example: Submarine fiber cables connecting continents.



e. Wi-Fi

Definition: A wireless technology that allows devices to connect to a local network without physical cables.

Features:

- ▶ *Based on IEEE 802.11 standards.*
- ▶ *Typical range: 20–50 meters indoors.*
- ▶ *Provides mobility and flexibility.*

Limitations:

- ▶ *Slower than wired connections.*
- ▶ *Sensitive to interference and security attacks.*

Example: Connecting a smartphone to the Internet in a coffee shop.

Comparison:



Technology	Main Role	Wired/Wireless	Example Use
Router	Connects networks, routes data	Both	Home router to access Internet
Switch	Connects devices in a LAN	Wired	Office connecting many PCs
Modem	Converts signals for ISP connection	Wired	DSL or cable Internet modem
Optical Fiber	High-speed medium for transmitting data	Wired	Internet backbone, ISP connections
Wi-Fi	Wireless access to a LAN and Internet	Wireless	Laptop or phone Internet connection

5.1.3 Peripherals

Peripherals are external devices connected to a computer. They allow input, output, or additional functionality.

a. Printers (Imprimantes)

- **Definition:** Output devices that produce a paper copy (hard copy) of digital documents.
- **Types:**
 - ▶ **Inkjet printers:** good for home use, print with liquid ink.
 - ▶ **Laser printers:** faster and better for offices, use toner powder.
 - ▶ **3D printers:** build three-dimensional objects layer by layer.
- **Use:** Printing documents, images, reports, designs.



b. Scanners

- **Definition:** Input devices that convert physical documents or images into digital format.
- **Types:**
 - ▶ **Flatbed scanners:** common for offices.
 - ▶ **Handheld scanners:** portable.
 - ▶ **Barcode scanners:** used in shops.
- **Use:** Digitizing paper documents, archiving photos, reading barcodes.



c. Keyboards (Claviers)

- **Definition:** Input devices that allow users to type text, numbers, and commands.
- **Types:**
 - ▶ **Standard QWERTY keyboards.**
 - ▶ **Virtual keyboards (on smartphones/tablets).**
 - ▶ **Specialized keyboards (gaming, ergonomic).**
- **Use:** Writing, programming, gaming, entering commands.



d. Mouse (Souris)

Definition: Pointing devices used to control a cursor on the screen.

Types:

- ▶ **Mechanical (ball-based, older).**
- ▶ **Optical (laser/LED, modern).**
- ▶ **Wireless (Bluetooth, RF).**

Use: Selecting, clicking, dragging, drawing, gaming.



e. Cameras (Caméras)

- **Definition:** Input devices that capture images and videos.
- **Types:**
 - ▶ *Webcams:* used for video calls and online meetings.
 - ▶ *Digital cameras:* higher quality photography.
 - ▶ *Surveillance cameras:* security monitoring.
- **Use:** Videoconferencing, online classes, security, photography.



f. Sensors (Capteurs)

- **Definition:** Input devices that detect physical changes in the environment and convert them into digital signals.
- **Examples:**
 - ▶ *Temperature sensors (thermometers).*
 - ▶ *Motion sensors (used in alarms).*
 - ▶ *Biometric sensors (fingerprint scanners, facial recognition).*
 - ▶ *IoT sensors (humidity, pressure, air quality).*
- **Use:** Smartphones, smart homes, medical devices, industrial automation.

Comparison:

Peripheral	Input/Output	Main Function	Example Use
Printer	Output	Produces hard copies	Printing reports
Scanner	Input	Digitizes paper documents	Scanning ID cards
Keyboard	Input	Text and command entry	Typing essays, coding
Mouse	Input	Cursor control and selection	Navigation, gaming
Camera	Input	Captures images and video	Zoom meetings, security
Sensor	Input	Detects physical/environment data	Smart devices, biometrics

5.1.4 Mobiles Dispositive

Definition: Mobile devices are portable electronic tools that allow users to access information, communicate, and perform digital tasks anywhere. They include smartphones, tablets, smartwatches, and other wireless gadgets. These devices combine computing power, internet

connectivity, and multimedia functions in a compact form. They use operating systems such as Android or iOS to run applications for messaging, browsing, navigation, and productivity. Their main characteristics include processing speed, memory capacity, battery life, display quality, and connectivity options like Wi-Fi or 5G. Mobile devices play a key role in modern life, supporting learning, work, and entertainment while enabling constant access to digital resources.

a. Smartphones

- **Definition:** Portable devices that combine the functions of a phone with those of a computer.
- **Features:** Touchscreen, operating systems (Android, iOS), internet access, cameras, GPS, sensors.
- **Use:** Communication (calls, messages), browsing, apps, banking, entertainment.
- **Difference:** More powerful and versatile than simple mobile phones.



b. Tablets

- **Definition:** Portable computers with a larger touchscreen than smartphones, designed mainly for multimedia, reading, and browsing.
- **Features:** Lightweight, long battery life, supports apps, stylus for note-taking/drawing.
- **Use:** E-learning, reading e-books, presentations, creative design.
- **Difference:** Larger screen than smartphones, but less powerful than laptops.



c. Assistants Numériques Personnels (PDA)

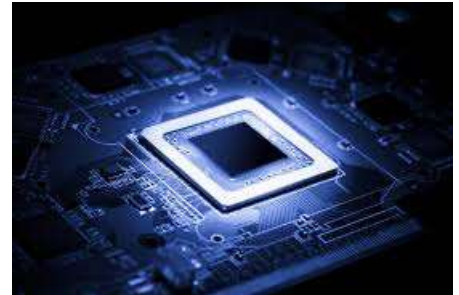
- **Definition:** Early handheld devices (before modern smartphones) used to manage personal information.
- **Features:** Small screen, stylus, limited storage, calendar, notes, simple apps.
- **Use:** Scheduling, contacts, notes.



- **Difference:** Replaced today by smartphones, which offer more advanced features.
- **Example:** Palm Pilot, HP iPAQ.

5.1.5 Smart Chips

Definition: Smart chips (**Puces intelligentes**) are **small electronic components** that add intelligence and connectivity to devices. They are designed to process information, communicate with other systems, and perform automated tasks without constant human control. They play a central role in modern technologies such as mobile phones, connected objects, and security systems.



- Smart chips are dynamic elements in Google Workspace documents (Docs, Sheets, Slides)
- They give quick access to related information and actions from other apps
- They can include contacts, files, dates, places, or financial data
- They turn a basic link into an interactive object with contextual data and actions

How smart chips work

- Type “@” in a document or spreadsheet
- Choose an element from the list such as contact, file, event, or place
- Hover over the chip to see extra information like file previews or event details
- Click the chip to open files, access pages, or trigger relevant actions

Examples

- Contact chip integrates a name or email for quick access to contact details
- File chip references Google Docs, Sheets, or Slides for fast access
- Event chip links to calendar event details
- Place chip embeds an address or location
- Finance chip displays stock, currency, or fund information

Benefits

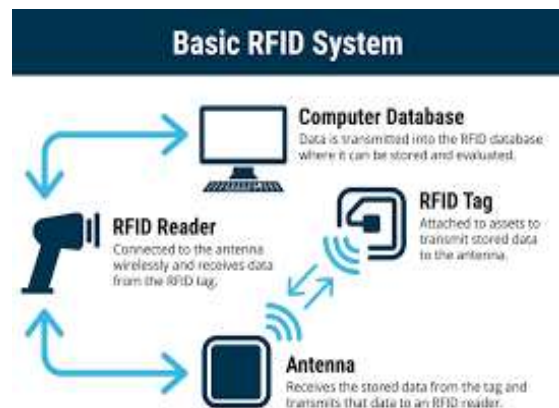
- Saves time by accessing information without leaving the document
- Improves collaboration by sharing and accessing project details in one place
- Enriches content by converting plain links into interactive elements

a. RFID (Radio Frequency Identification)

Definition: It is a technology used to store and retrieve data remotely through radio waves without direct visual contact. It relies on a system that includes an RFID tag containing a microchip and an antenna, and a reader that interrogates the tag. RFID technology is used in multiple sectors for object identification, inventory management, access control, and tracking of people or animals. It raises privacy concerns.

How it works

- **RFID Tag:** contains stored data and an antenna
- **RFID Reader:** emits radio waves to power the tag and collect the data it returns
- **Data Transmission:** the tag uses its antenna to send information back to the reader through radio waves



Common applications

- Logistics and inventory management for rapid product identification and tracking (Finkenzeller, 2010)
- Access control using badges or embedded chips to enter buildings or systems (Want, 2006)
- Animal tracking through implanted RFID microchips for identification and monitoring (Hodges et al., 2007)
- Asset management including clothing, personal protective equipment, or surgical tools (Dobkin, 2013)

Ethical considerations and limitations

- **Privacy:** the discreet implantation of RFID chips and the ability to track individuals raise ethical issues (Rotter, 2008)
- **Data theft protection:** RFID-blocking materials are now used in wallets and bags to prevent unauthorized scanning of banking data (Juels, 2006)

Frequency types

- **Low Frequency (LF):** mainly used for animal identification such as pets and livestock (Finkenzeller, 2010)

- **High Frequency (HF):** often used for access control in buildings and elevators (Dobkin, 2013)
- **Ultra High Frequency (UHF):** enables longer read ranges and is used for inventory and object tracking (Want, 2006)

b. NFC stands for Near Field Communication

A short-range wireless technology enables two electronic devices to exchange data when they are within a few centimeters. It builds on RFID principles to create a secure, quick link. NFC is used for contactless payments, data sharing, and fast device pairing

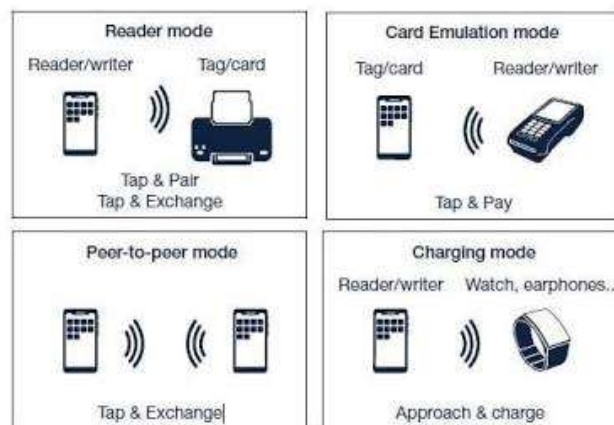


How NFC works

- ▶ Short-range communication uses radio waves at 13.56 MHz over distances up to 4 cm
- ▶ Inductive coupling happens when one device generates an electromagnetic field that induces current in the other, creating a link

Three modes of operation:

- ▶ **Peer-to-Peer** for exchanging data between two NFC-enabled devices
- ▶ **Read/Write** for interacting with passive NFC tags through an active device
- ▶ **Card Emulation** for using the device like a smart card for payments or access



Common uses

- Contactless payments through Google Pay, Samsung Pay, and similar services
- Data transfer for contacts, photos, or links by tapping two devices
- Device pairing for headphones or speakers without manual Bluetooth setup

- Access control for public transport, buildings, or hotel rooms with phones or NFC cards
- Smart home automation by using NFC tags with platforms like Home Assistant or Apple HomeKit

c. Embedded Processors (processeurs embarqués)

Definition: An embedded processor is the central component of an embedded system, specialized computer designed to perform a specific task within a larger device such as a smartphone, car, or connected thermostat. These processors are optimized for cost, size, and power constraints. They are usually based on low-power microcontrollers or microprocessors that integrate a processor, memory, and input/output interfaces on a single chip

Characteristics and roles

- **Specialization:** designed for a single, well-defined function, which increases efficiency
- **Integration:** embedded directly into the final product and works with embedded software to control specific functions
- **Constraints:** strict requirements for cost, energy consumption, size (small form factor), and real-time response
- **Components:** typically include a microcontroller with processor, memory, and I/O, plus sensors and actuators



Applications

- **Automotive:** dashboard control, parking sensors, and communication between vehicle subsystems
- **Home automation:** lighting systems, shutters, and connected voice assistants
- **Industrial:** process automation, robot control, and production line management
- **Medical:** digital thermometers, blood pressure monitors, and other medical devices
- **Everyday devices:** pacemakers, smart thermostats, and connected objects

Examples of manufacturers

- AMD provides x86 CPU and APU solutions for embedded systems that require higher performance and graphics or AI capabilities with low power use.
- Intel, NXP, and STMicroelectronics are major producers of embedded processors and microcontrollers for a wide range of applications.

d. IoT Sensors (capteurs IoT)

Definition: An IoT sensor is an electronic device that measures physical or environmental data such as temperature, humidity, or movement. It transmits this data over a connected network to a platform for analysis and action. It enables remote monitoring, task automation, and data-driven decision-making. IoT sensors play a central role in areas like precision agriculture, building management, and cold chain monitoring.



How it works

- ▶ **Measurement:** the sensor detects and measures specific environmental parameters
- ▶ **Conversion:** physical signals are converted into digital data
- ▶ **Transmission:** data is sent through wireless connections such as WiFi, LoRa, 4G, or Bluetooth to an IoT platform
- ▶ **Analysis and action:** the platform processes the data to trigger alerts, predict needs, automate tasks, or launch corrective actions

Common applications

- **Precision agriculture:** monitoring soil moisture and nutrients to optimize irrigation and fertilization.
- **Energy and building management:** optimizing HVAC systems to reduce energy consumption.
- **Cold chain monitoring:** maintaining required temperatures for food and pharmaceutical storage.
- **Healthcare:** tracking body parameters to detect anomalies or predict cycles.
- **Industry:** monitoring fuel levels, gas presence, or other variables to prevent shortages or incidents.

Benefits

- Informed decisions through real-time data access
- Automation of tasks and processes
- Operational efficiency and cost reduction
- Remote monitoring of environments and systems